

**대곡교  
수위 예측 AI  
성능 개선 결과 3**  
( 시계열 )

**※ 강우 이벤트 전부 학습으로 사용**

# MIET 학습 데이터 기본 설정

- 관심 수위 : 궁내교 ( 2 ) 대곡교 ( 3.8 )
- smoothing : 궁내교 ( 1 ) 대곡교 ( 2 )
- learning\_rate = 0.0001
- Loss\_function = MSE
- Seq\_length = 36 ( 6시간 )
- Num\_of\_layer = 12 ( 각 LSTM 3개 처리 )

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출

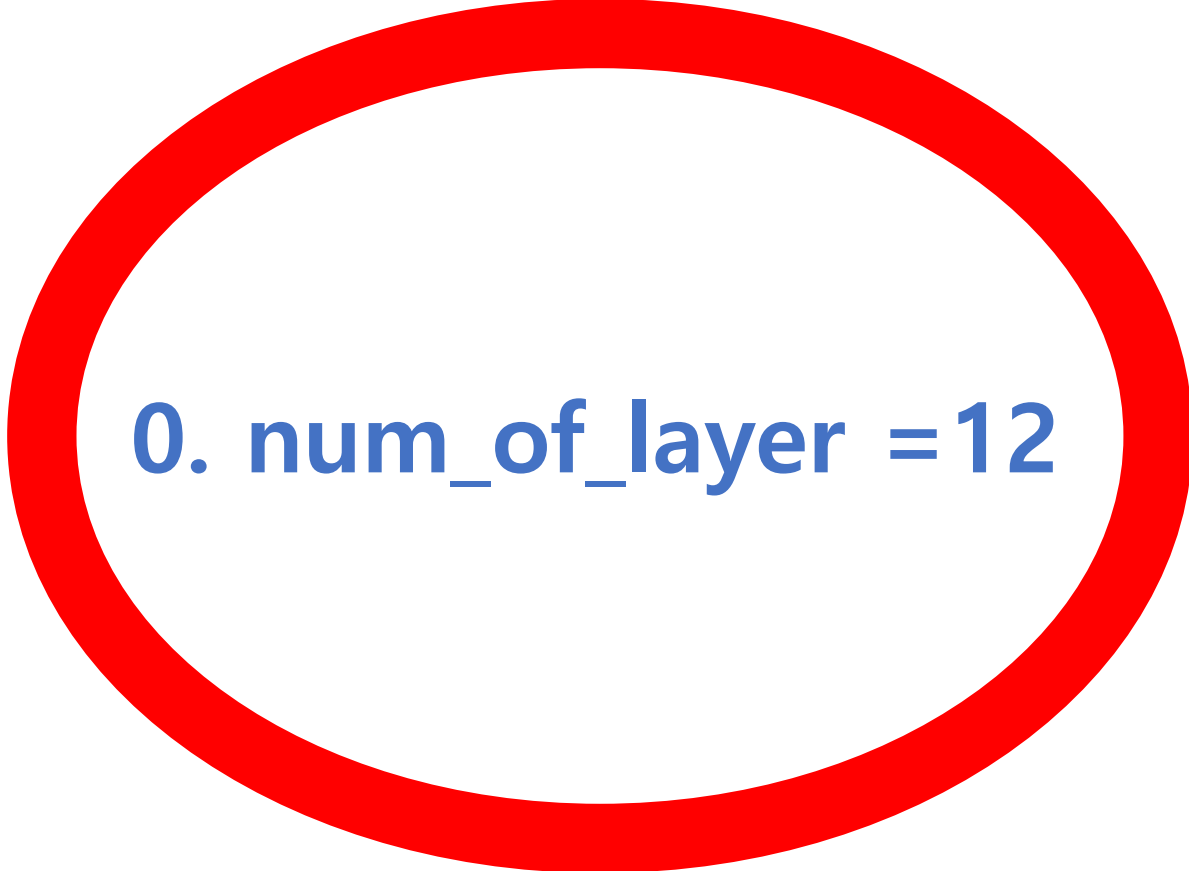
| time                | 서울시<br>(대곡교) | 성남시<br>(성남북초교) | 광주시<br>(남한산초교) | 성남시<br>(대장동) | 성남시<br>(구미초교) | 성남시<br>(한국학중앙연구원) | 성남시<br>(궁내교)_WL | 성남시<br>(궁내교)_Q | 서울시<br>(대곡교)_WL | 서울시<br>(대곡교)_Q | 대곡교_Ti | 궁내교_Ti |
|---------------------|--------------|----------------|----------------|--------------|---------------|-------------------|-----------------|----------------|-----------------|----------------|--------|--------|
| 2014-07-24 15:30:00 | 0.0          | 0.0            | 0.0            | 0.0          | 0.0           | 0.0               | 1.12            | 5.11           | 2.17            | 40.56          | 0.0    | 0.0    |

학습 : 총 1420개 中 1412개 데이터 파일 학습데이터로 사용 ( 결측 데이터 5개 )

검증 : 학습 데이터에서 제외한 3개 데이터 사용 ++ 학습에서 검증에 사용된 3개 데이터 사용

- 학습 : ~ 2022 68번 ( 988개 )
- 검증 : ~ 2024 96번 ( 282개 )
- 테스트 : ~ 2025 109번 ( 142개 )

**seq\_length = 36**  
**+ num\_of\_layer 조정**

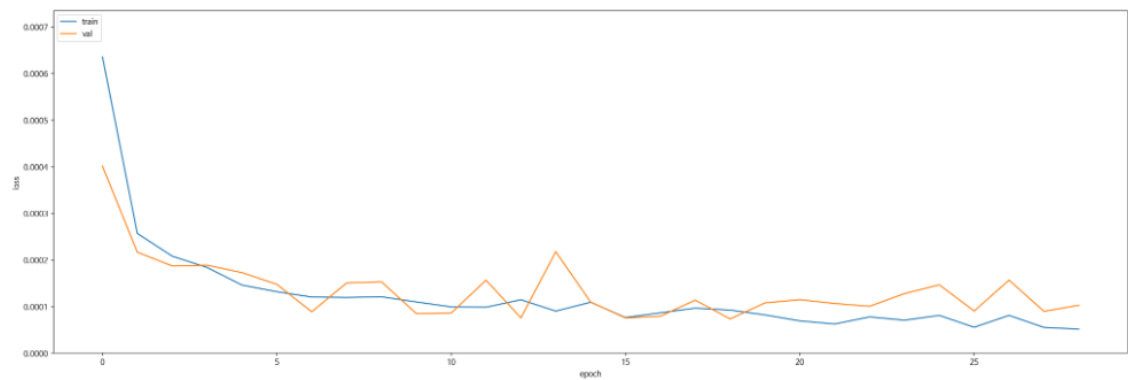


**0. num\_of\_layer = 12**

num\_of\_layer =12  
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

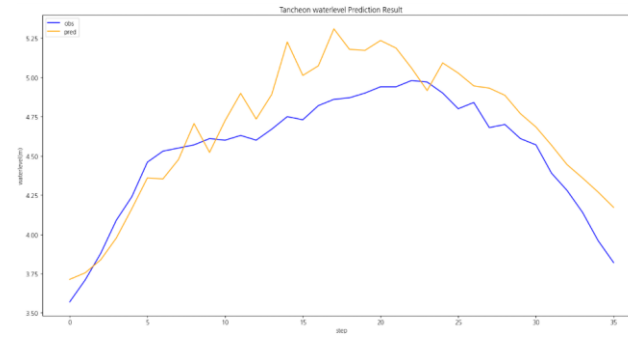
| Train<br>(23111 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|--------------------|--------|--------|--------|----------|----------|----------|
|                    | 0.0348 | 0.0167 | 0.0545 | 0.977872 | 0.981024 | 1.489537 |

| Valid<br>(6282 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|-------------------|--------|--------|--------|----------|----------|----------|
|                   | 0.0314 | 0.0141 | 0.0533 | 0.970796 | 0.973404 | 0.382375 |

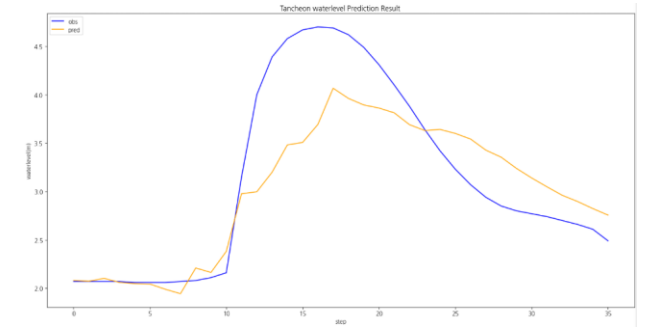
| Test<br>(5472 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|------------------|--------|--------|--------|----------|----------|----------|
|                  | 0.0297 | 0.0136 | 0.0621 | 0.945628 | 0.946119 | 5.689971 |

num\_of\_layer =12  
예측 코드 - 대곡교

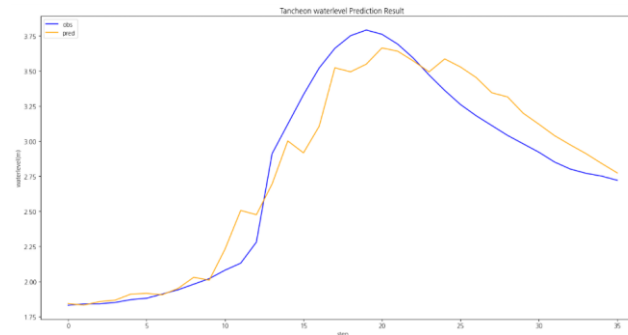
|      | 대곡교                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| 증설 전 | testdata_220712<br>r2 : 0.6571355333013245<br>mse : 0.04801457060069652<br>mae : 0.19200435611936775<br>rmse : 0.21912227317344196 |
|      | testdata_230711<br>r2 : 0.7198263671371292<br>mse : 0.2550023667304158<br>mae : 0.3688772605525123<br>rmse : 0.5049775903249726    |
| 증설 후 | testdata_250716<br>r2 : 0.9231141326275117<br>mse : 0.03550482569334603<br>mae : 0.14576794703801474<br>rmse : 0.18842724243947856 |
|      | testdata_250814<br>r2 : 0.8274971285050222<br>mse : 0.059875573660618985<br>mae : 0.213871206442515<br>rmse : 0.24469485826355034  |



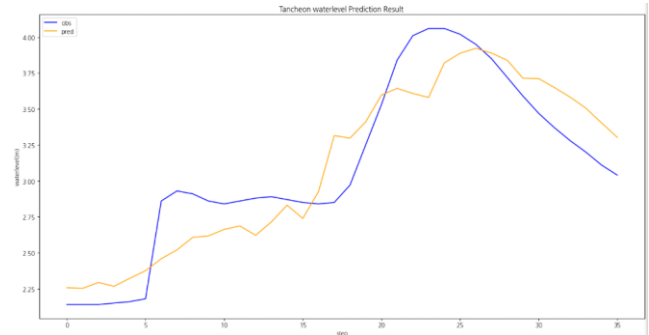
testdata\_220712



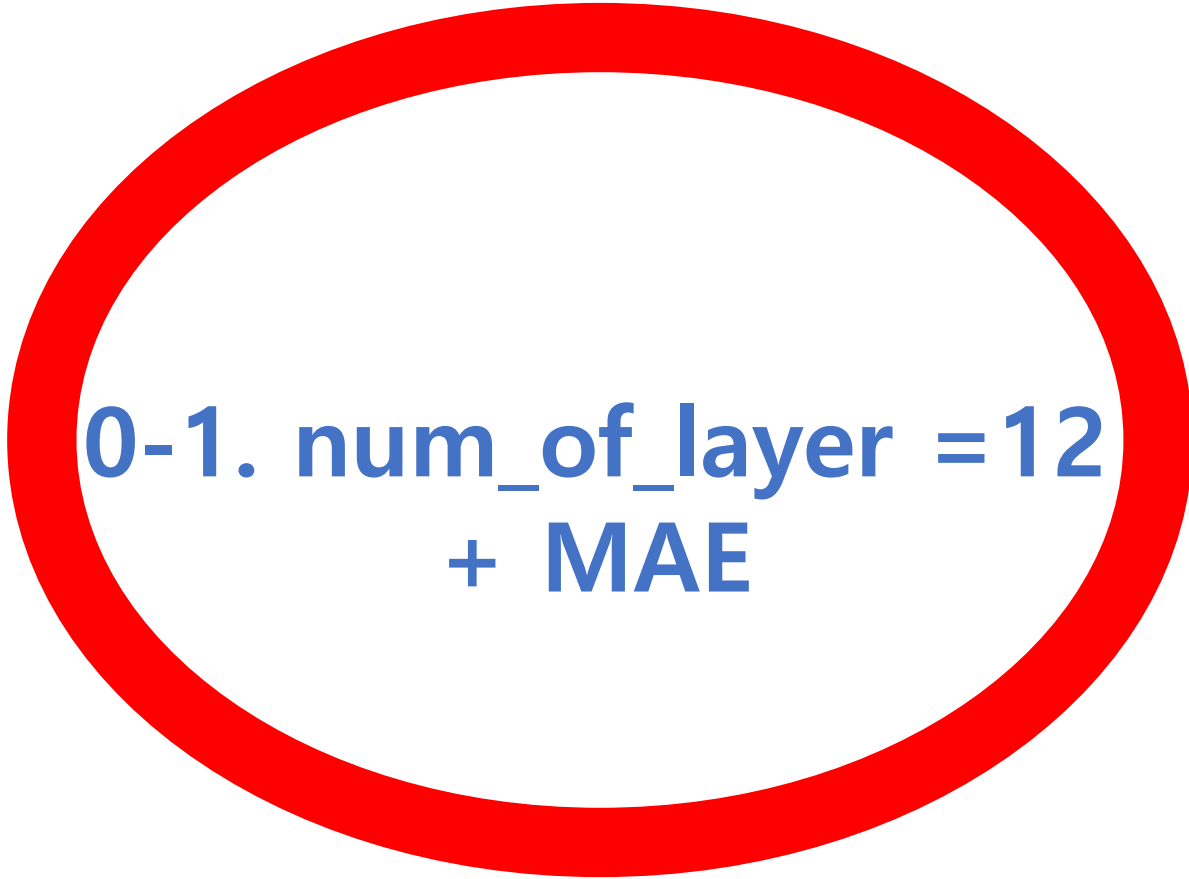
testdata\_230711



testdata\_250716



testdata\_250814



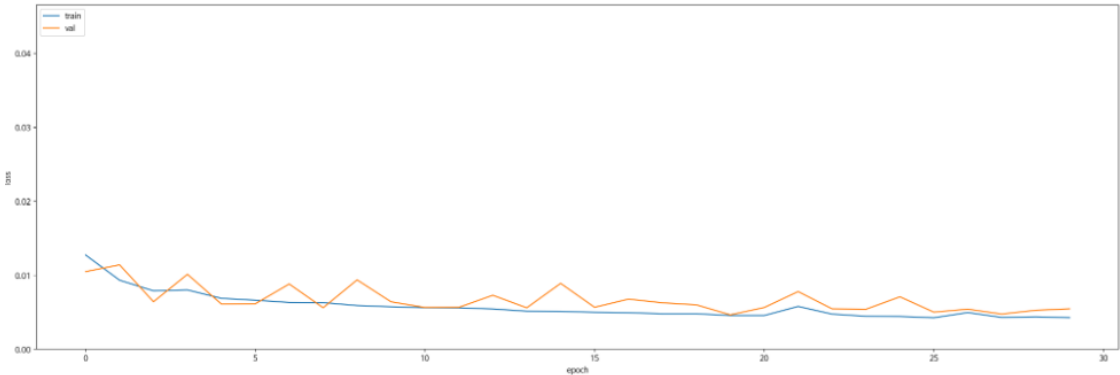
**0-1. num\_of\_layer = 12  
+ MAE**



num\_of\_layer =12  
loss\_function = MAE  
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

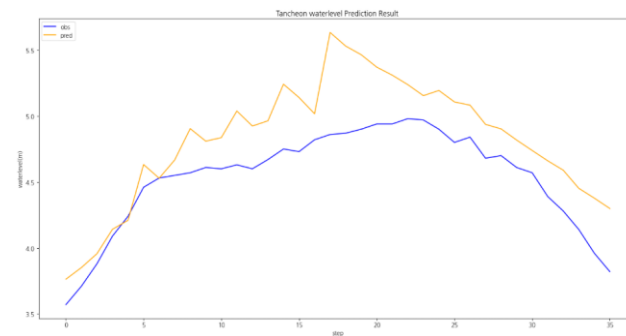
| Train<br>(23111 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|--------------------|--------|--------|--------|----------|----------|----------|
|                    | 0.0311 | 0.0144 | 0.0599 | 0.973250 | 0.975980 | 1.864916 |

| Valid<br>(6282 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|-------------------|--------|--------|--------|----------|----------|----------|
|                   | 0.0289 | 0.0129 | 0.0543 | 0.969639 | 0.970755 | 0.981632 |

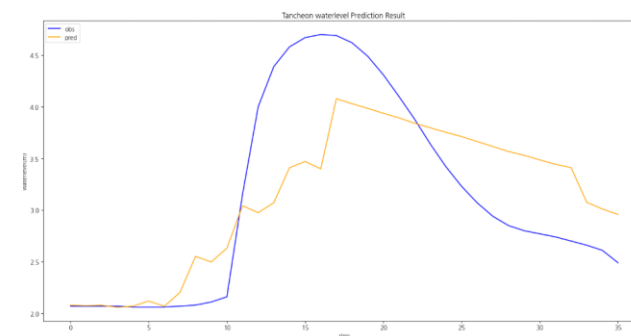
| Test<br>(5472 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|------------------|--------|--------|--------|----------|----------|----------|
|                  | 0.0271 | 0.0120 | 0.0682 | 0.934381 | 0.936408 | 5.370354 |

num\_of\_layer =12  
 loss\_function = MAE  
 예측 코드 - 대곡교

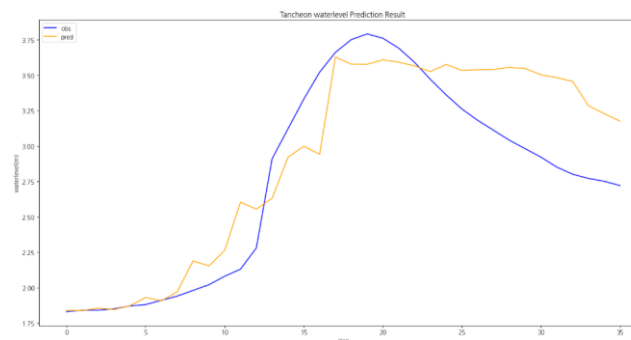
|      | 대곡교                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 증설 전 | testdata_220712<br>r2 : 0.2091296404860017<br>mse : 0.11075309459307643<br>mae : 0.28849312014049944<br>rmse : 0.3327958752645177 |
|      | testdata_230711<br>r2 : 0.5920753404002524<br>mse : 0.3712760283068737<br>mae : 0.47554510619905266<br>rmse : 0.6093242390606775  |
| 증설 후 | testdata_250716<br>r2 : 0.7620351947859629<br>mse : 0.10988884198110321<br>mae : 0.25544776280721027<br>rmse : 0.331494859660151  |
|      | testdata_250814<br>r2 : 0.5902563975559536<br>mse : 0.14222159340008522<br>mae : 0.32031501584582855<br>rmse : 0.377122782923659  |



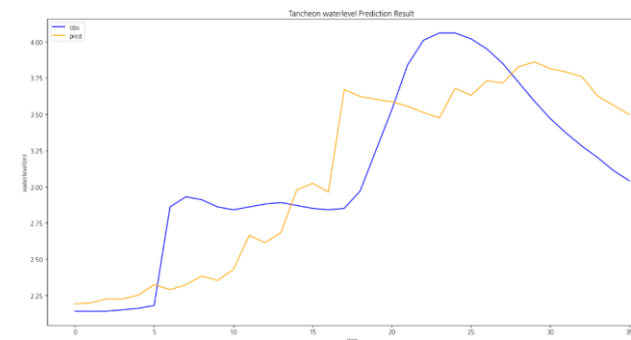
testdata\_220712



testdata\_230711



testdata\_250716



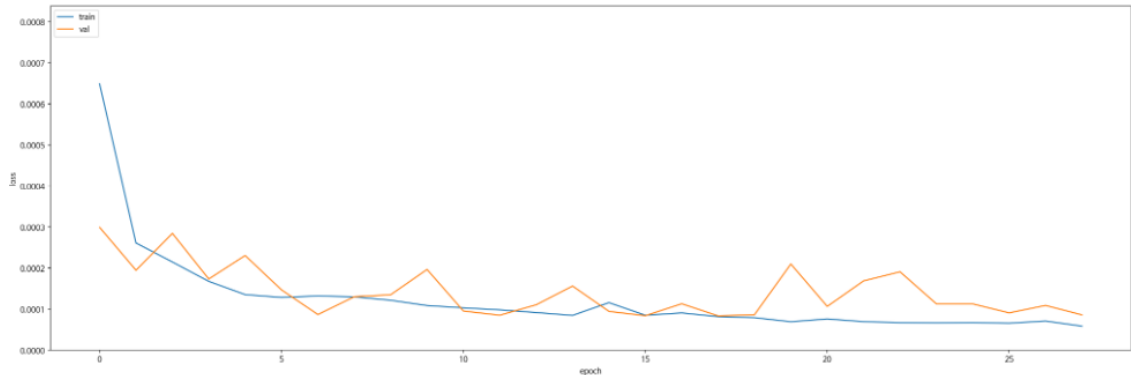
testdata\_250814

**1. num\_of\_layer = 18**

num\_of\_layer =18  
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

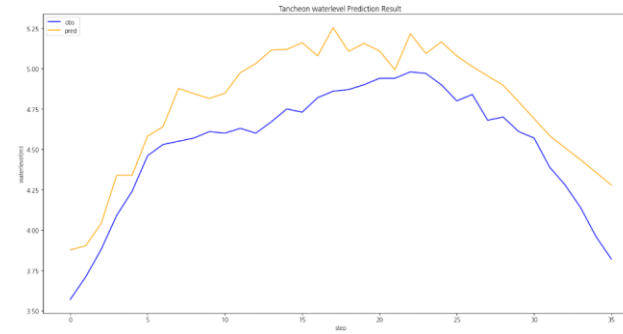
| Train<br>(23111 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|--------------------|--------|--------|--------|----------|----------|----------|
|                    | 0.0384 | 0.0182 | 0.0604 | 0.972778 | 0.974865 | 4.641500 |

| Valid<br>(6282 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|-------------------|--------|--------|--------|----------|----------|----------|
|                   | 0.0376 | 0.0174 | 0.0569 | 0.966656 | 0.967987 | 2.900384 |

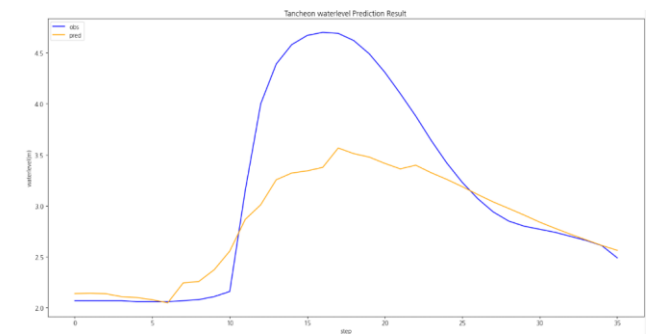
| Test<br>(5472 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|------------------|--------|--------|--------|----------|----------|----------|
|                  | 0.0363 | 0.0169 | 0.0677 | 0.935354 | 0.937525 | 8.181360 |

num\_of\_layer =18  
예측 코드 - 대곡교

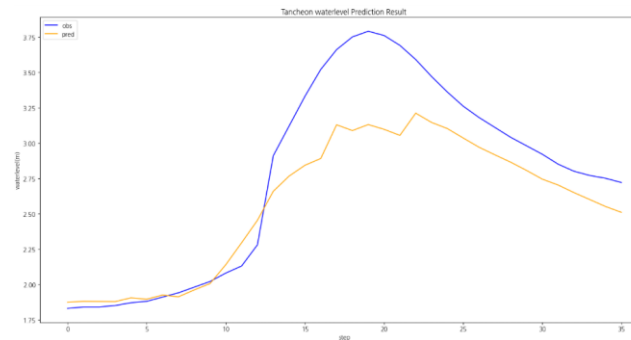
|      | 대곡교                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 증설 전 | testdata_220712<br>r2 : 0.46494493229205325<br>mse : 0.0749288474570961<br>mae : 0.253041501575046<br>rmse : 0.27373134175153585  |
|      | testdata_230711<br>r2 : 0.6025684964802238<br>mse : 0.36172559485772043<br>mae : 0.3919211355845134<br>rmse : 0.6014362766392799  |
| 증설 후 | testdata_250716<br>r2 : 0.7837896284275703<br>mse : 0.09984294667032062<br>mae : 0.23748457458284167<br>rmse : 0.3159793453223179 |
|      | testdata_250814<br>r2 : 0.6523937960110676<br>mse : 0.12065376472549574<br>mae : 0.27478178872002496<br>rmse : 0.3473525078727599 |



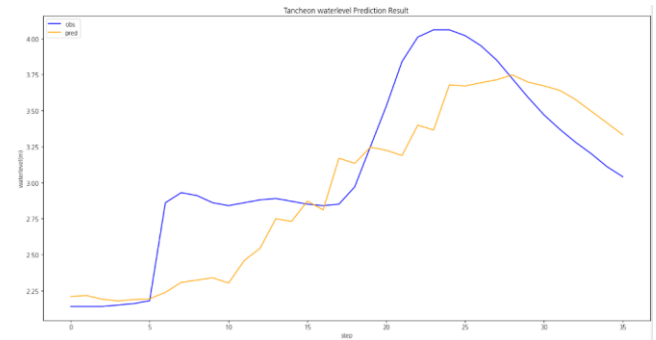
testdata\_220712



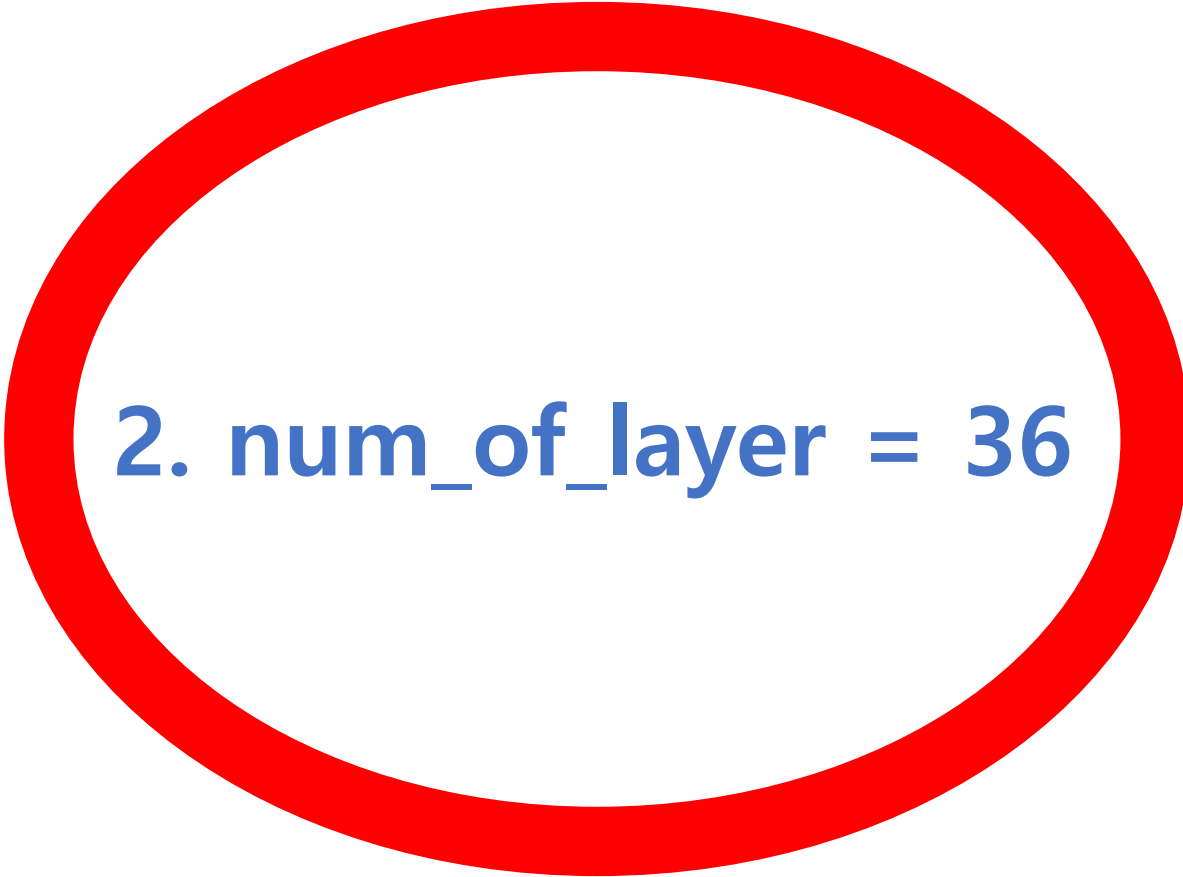
testdata\_230711



testdata\_250716



testdata\_250814

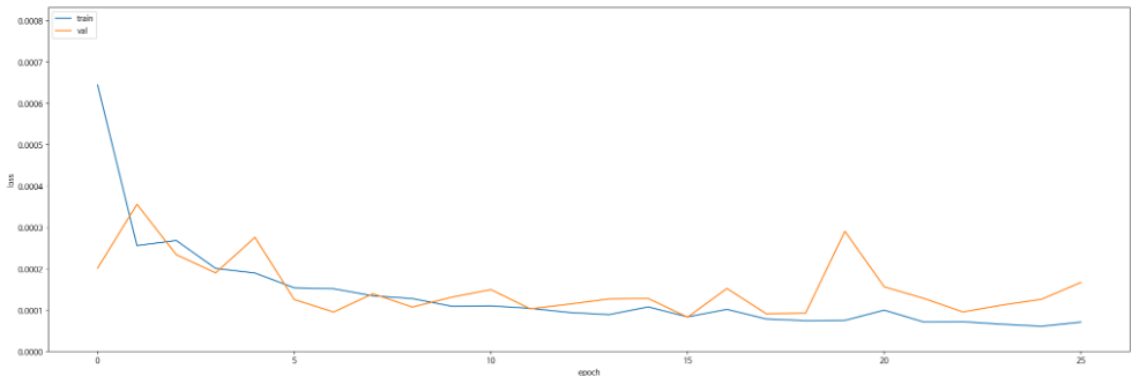


**2. num\_of\_layer = 36**

num\_of\_layer =36  
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

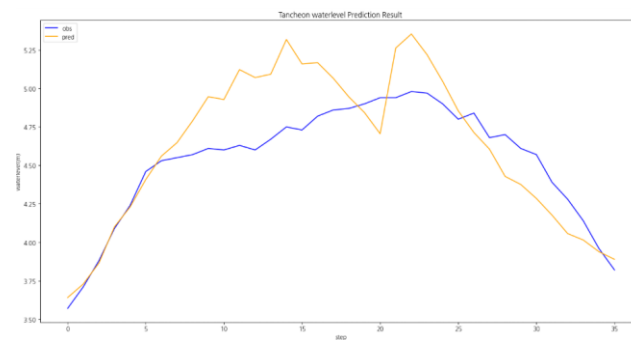
| Train<br>(23111 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|--------------------|--------|--------|--------|----------|----------|----------|
|                    | 0.0337 | 0.0160 | 0.0561 | 0.976522 | 0.978192 | 4.313771 |

| Valid<br>(6282 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|-------------------|--------|--------|--------|----------|----------|----------|
|                   | 0.0358 | 0.0163 | 0.0567 | 0.966926 | 0.968519 | 3.448825 |

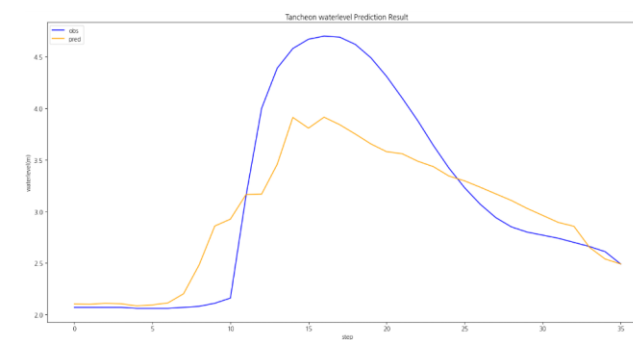
| Test<br>(5472 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|------------------|--------|--------|--------|----------|----------|----------|
|                  | 0.0314 | 0.0144 | 0.0628 | 0.944348 | 0.945049 | 6.891850 |

num\_of\_layer =36  
예측 코드 - 대곡교

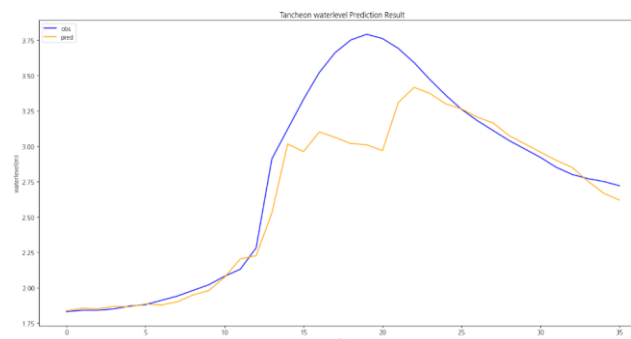
|      | 대곡교                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 증설 전 | testdata_220712<br>r2 : 0.5347686030011513<br>mse : 0.06515077509181906<br>mae : 0.2023961880471971<br>rmse : 0.2552464986866991  |
|      | testdata_230711<br>r2 : 0.749784161153433<br>mse : 0.22773603085819336<br>mae : 0.34499009158876204<br>rmse : 0.47721696413496595 |
| 증설 후 | testdata_250716<br>r2 : 0.8294799701303481<br>mse : 0.07874378146005706<br>mae : 0.1590827772352431<br>rmse : 0.28061322395791877 |
|      | testdata_250814<br>r2 : 0.7660424705680733<br>mse : 0.08120642378620117<br>mae : 0.24448778284920586<br>rmse : 0.2849674082876868 |



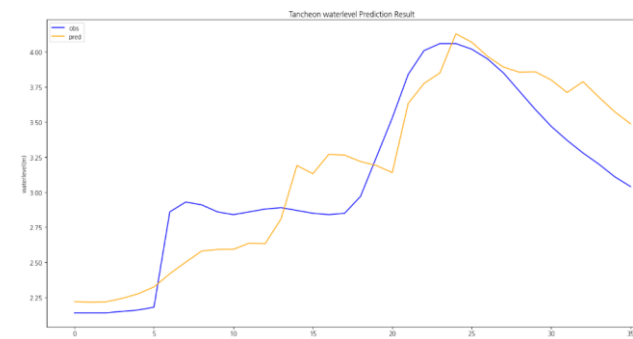
testdata\_220712



testdata\_230711



testdata\_250716



testdata\_250814



**seq\_length = 36 +  
num\_of\_layer = 12 이 제일 나옴**

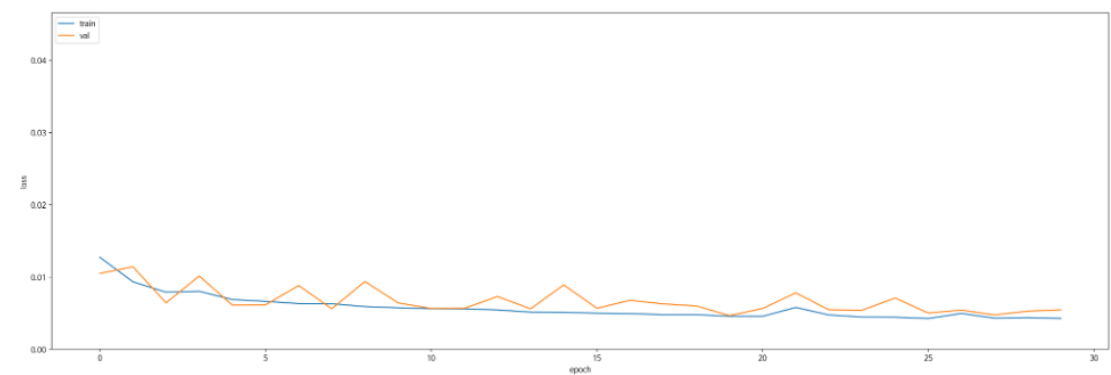
**Loss\_function = MAE**  
**Learning\_rate = 0.0005**

**1. num\_of\_layer = 12**

num\_of\_layer = 12  
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

| Train<br>(23111 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|--------------------|--------|--------|--------|----------|----------|----------|
|                    | 0.0311 | 0.0144 | 0.0599 | 0.973250 | 0.975980 | 1.864916 |

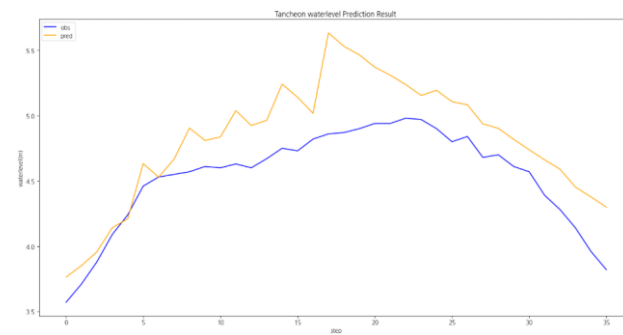
| Valid<br>(6828 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|-------------------|--------|--------|--------|----------|----------|----------|
|                   | 0.0289 | 0.0129 | 0.0543 | 0.969639 | 0.970755 | 0.981632 |

| Test<br>(5472 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|------------------|--------|--------|--------|----------|----------|----------|
|                  | 0.0271 | 0.0120 | 0.0682 | 0.934381 | 0.936408 | 5.370354 |

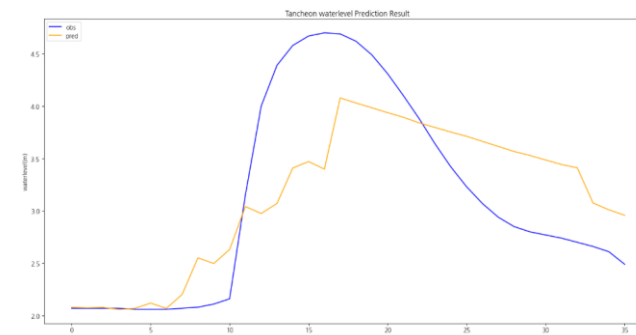
num\_of\_layer = 12

예측 코드 - 대곡교

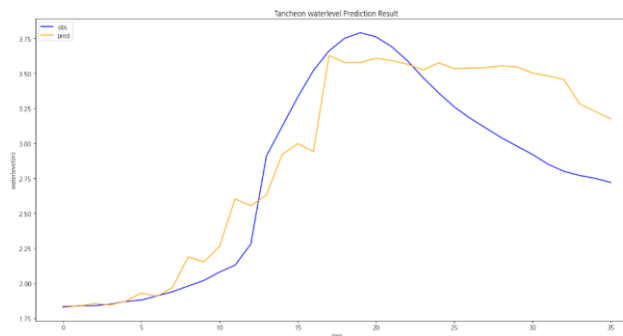
|      | 대곡교                                                                                                                               |
|------|-----------------------------------------------------------------------------------------------------------------------------------|
| 증설 전 | testdata_220712<br>r2 : 0.2091296404860017<br>mse : 0.11075309459307643<br>mae : 0.28849312014049944<br>rmse : 0.3327958752645177 |
|      | testdata_230711<br>r2 : 0.5920753404002524<br>mse : 0.3712760283068737<br>mae : 0.47554510619905266<br>rmse : 0.6093242390606775  |
| 증설 후 | testdata_250716<br>r2 : 0.7620351947859629<br>mse : 0.10988884198110321<br>mae : 0.25544776280721027<br>rmse : 0.331494859660151  |
|      | testdata_250814<br>r2 : 0.5902563975559536<br>mse : 0.14222159340008522<br>mae : 0.32031501584582855<br>rmse : 0.377122782923659  |



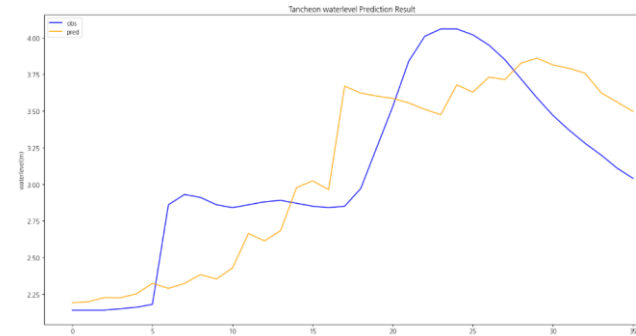
testdata\_220712



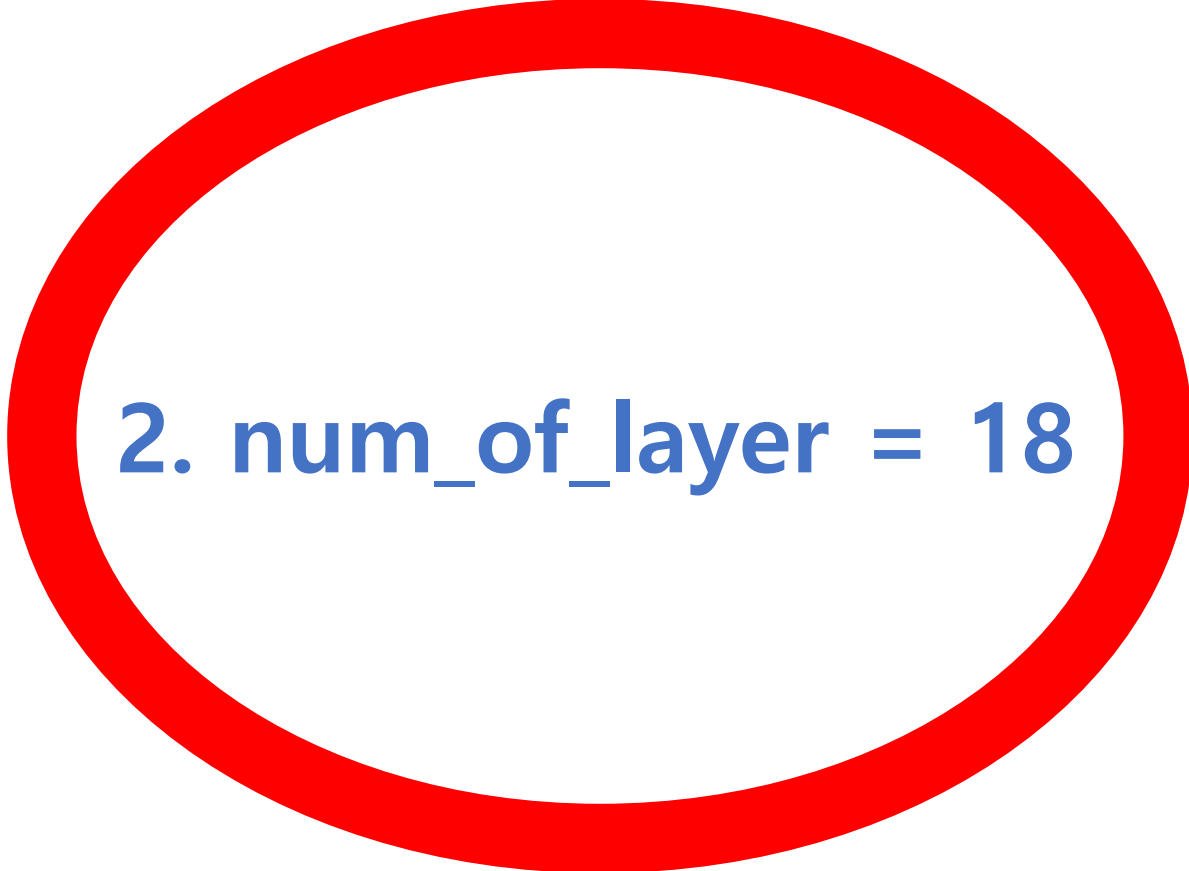
testdata\_230711



testdata\_250716



testdata\_250814

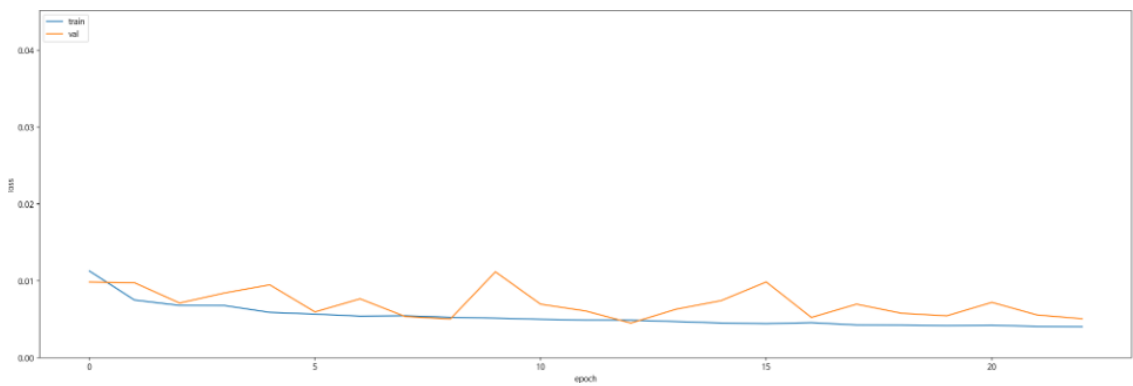


**2. num\_of\_layer = 18**

num\_of\_layer = 18  
학습 코드 - 대곡교

Train : Valid : Test = 7 : 2 : 1 비율로 학습 진행

Train Chart



일괄 예측 수행 결과

| Train<br>(23111 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER       |
|--------------------|--------|--------|--------|----------|----------|-----------|
|                    | 0.0319 | 0.0145 | 0.0653 | 0.968229 | 0.970063 | 13.060085 |

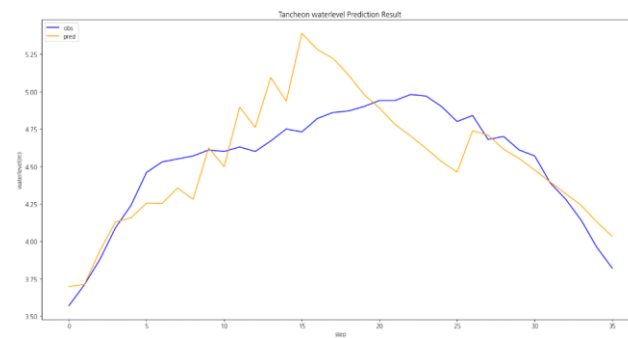
| Valid<br>(6828 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|-------------------|--------|--------|--------|----------|----------|----------|
|                   | 0.0281 | 0.0124 | 0.0555 | 0.968298 | 0.969206 | 1.881732 |

| Test<br>(5472 건) | MAE    | MAPE   | RMSE   | NSE      | CoR2     | QER      |
|------------------|--------|--------|--------|----------|----------|----------|
|                  | 0.0267 | 0.0118 | 0.0682 | 0.934294 | 0.935146 | 3.396203 |

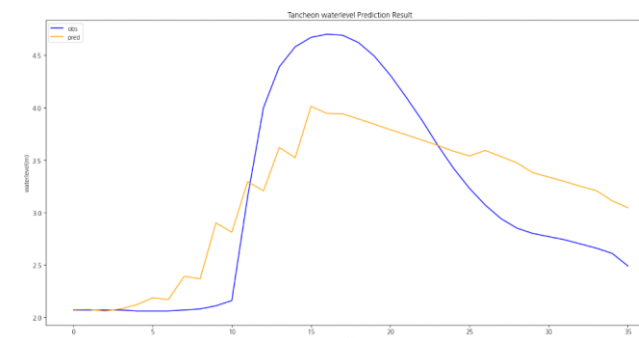
num\_of\_layer = 18

예측 코드 - 대곡교

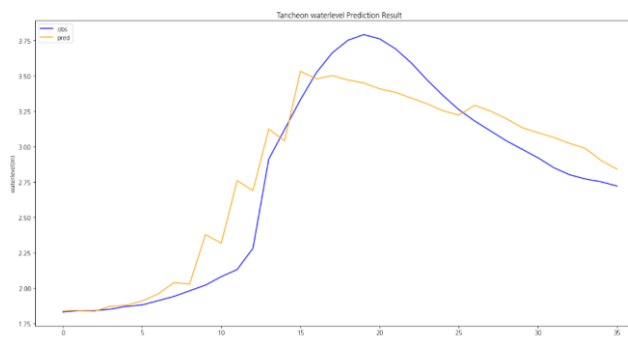
|      | 대곡교                                                                                                                                |
|------|------------------------------------------------------------------------------------------------------------------------------------|
| 증설 전 | testdata_220712<br>r2 : 0.5988864025277925<br>mse : 0.05617175010921909<br>mae : 0.18513488743040302<br>rmse : 0.23700580184716807 |
|      | testdata_230711<br>r2 : 0.6995342686744681<br>mse : 0.27347138924702724<br>mae : 0.43939690192540487<br>rmse : 0.5229449198979059  |
| 증설 후 | testdata_250716<br>r2 : 0.8985794355496304<br>mse : 0.04683460804422999<br>mae : 0.16922600640190966<br>rmse : 0.2164130496163066  |
|      | testdata_250814<br>r2 : 0.6443384065170401<br>mse : 0.12344978233862086<br>mae : 0.2827947984801398<br>rmse : 0.3513542120689901   |



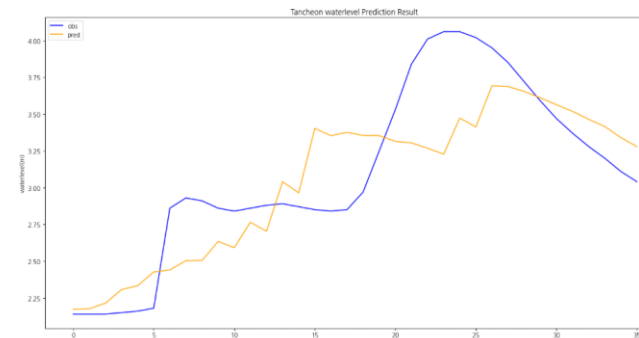
testdata\_220712



testdata\_230711



testdata\_250716



testdata\_250814



**Loss\_function = MAE**  
**Learning\_rate = 0.0005 은**  
**num\_of\_layer = 18 이 제일 나옴**

## 결론

- 관심 수위 : 궁내교 ( 2 ) 대곡교 ( 3.8 )
- smoothing : 궁내교 ( 1 ) 대곡교 ( 2 )
- learning\_rate = 0.0001
- Loss\_function = MSE
- Seq\_length = 36 ( 6시간 )
- Num\_of\_layer = 12 ( 각 LSTM 3개 처리 )

**num\_of\_layer = 12**